VISHAL DHAWAL

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Computer Science undergraduate specializing in Computer Vision and Deep Learning with experience spanning aerial image analysis, threat intelligence, and vision-based Human-Computer Interaction, using PyTorch, OpenCV, EfficientNet, and MediaPipe.

EDUCATION

- **Indian Institute of Information Technology Kottayam, India**
  Bachelor of Technology - Computer Science and Engineering; Aug 2023 - April 2027
  GPA: 7.3/10 (through Sem 6)

- **Indian Institute of Technology Mandi, India**
  Minor in Computer Science and Advanced Technologies; Oct 2024 - Oct 2025
  GPA: 7.4/10

PROJECTS

TeleWire Telegram SOCMINT • Threat Intelligence • NLP • Telethon — July 2026, **Github**

- Developed an AI-powered Telegram threat intelligence platform for automated monitoring, entity extraction, semantic threat classification, and cybercrime intelligence generation across public channels.
- Automated extraction of entities, semantic threat classification, OCR, and composite risk scoring to assist intelligence-driven cyber investigations.
- Built interactive investigation dashboards with campaign clustering, relationship visualization, and case reporting to support efficient analyst decision-making.

Aerial GCP Pose Computer Vision • PyTorch • EfficientNet B3 • Multi-Task learning — June 2026, **Github**

- Built a multi-task EfficientNet-B3 pipeline for GCP localization and shape classification from 4096×3000 aerial imagery using crop-based training and keypoint-aware augmentation.
- Designed a leakage-safe training and tiled inference pipeline with 512×512 crops and GroupKFold grouping by project/survey/GCP.
- Achieved 20.32 px mean localization error, ~100% shape classification accuracy, and ~1.0 Macro-F1, with inference successfully executed across 300 unlabeled test images.

VisionControl Human Computer Interaction • OpenCV • MediaPipe • PyAutoGUI — December 2025, **Github**

- Implemented a vision-based gesture system using OpenCV and MediaPipe to smoothly control cursor, clicks, volume, and brightness in real time with high accuracy.
- Achieved 92% gesture-tracking accuracy and 85% blink-click reliability by optimizing hand + face landmark processing.
- Improved interaction stability by 40% using smoothing + motion-filtering, resulting in smoother and more responsive control.

EXPERIENCE

AI/ML Intern — Cyber Crime Headquarters, Uttar Pradesh Police (June 2026 - July 2026)

- Developed CYBORG, an AI-powered investigation assistant leveraging LLMs, semantic retrieval, and document intelligence to assist cybercrime investigations through intelligent knowledge retrieval and contextual analysis.
- Designed AI workflows for knowledge retrieval and decision support while collaborating with senior law enforcement officials under strict operational confidentiality.

Mentor, MindQuest Mental Wellness Club – IIIT Kottayam (Sept 2025 – Aug 2026)

- Mentored volunteers through interviews, the GameOn event (300+ participants), and student engagement initiatives
- Contributed to planning, organizing, and executing mental wellness programs to strengthen student outreach and participation.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, SQL
- **Machine Learning & AI:** CNNs, NLP, Transformers (BERT), LLMs, Retrieval-Augmented Generation (RAG)
- **Computer Vision:** Hand Tracking, Face Mesh, Eye Tracking, Gesture Recognition, OpenCV, MediaPipe
- **Tools & Platforms:** Git & GitHub, FastAPI, WebSocket, VS Code, Jupyter Notebook, Google Colab, Firebase
- **Data Analytics:** Pandas, NumPy, Matplotlib, Data Visualization, Statistical Analysis, EDA
- **Interests:** Computer Vision, Human Computer Interaction, Applied Machine Learning, Hyperspectral Image Processing

CERTIFICATION

- OpenCV BootCamp — March 2025, **Certificates**
- ChatGPT Prompt Engineering for Developers — March 2025, **Certificates**